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Cochrane reviews – in their own words

Vitamin A supplementation for preventing morbidity and mortality in children from 6 months to 5 years of age

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"Vitamin A is important for the immune system. It helps to fight infection. When people don't get enough, they become vulnerable to diseases like diarrhoea and measles. For children in low and middle income countries, these can lead to death. The World Health Organisation estimates that 190 million children under the age of 5 don't get enough Vitamin A to stay healthy.

In high income countries, most people get Vitamin A through fresh foods including dairy products, oily fish, and orange vegetables like carrots and sweet potatoes. Vitamin A can also be added to foods like margarine. But in many low income countries, people don't get enough food or don't have access to a rich variety of foods. Supplements might be one way to overcome this.

We conducted this review (1) to provide a reliable estimate of the effects of vitamin A supplements for children between 6 months and 5 years of age. We found 43 clinical trials, in which some children received vitamin A while others had no intervention or a placebo. The trials took place in 19 countries, mostly in Asia. Except two that were conducted in Australia, these were all conducted in low and middle income countries. The oldest study was reported in 1976, but about half the studies were conducted in the last 15 years. In total, about 215,000 children took part in the research.

Children were about 2.5 years old when they were recruited, and they were followed for about 1 year, during which time children who received vitamin A were less likely to have an episode of measles or diarrhoea. We also found that after 1 year, children in the Vitamin A group were 24% less likely to have died than children who received a placebo. Specifically, they were less likely to die from diarrhoea and measles.

There were some differences in the size of the benefits across trials, but even if Vitamin A reduces child deaths by only 10%, this would be a large and important effect. In high risk areas, our findings mean that Vitamin A could reduce the number of deaths from 9 to 7 for every 100 children. If Vitamin A supplements were given to all children who don't get enough at the moment, this could prevent nearly 1 million deaths every year.

Although most of the large studies did not measure or report side effects, a few small studies did examine these during the first 48 hours. They found that among every 100 children who received Vitamin A rather than placebo, 4 more vomited; but smaller doses of vitamin A may be easier to tolerate than the extremely large doses used in these studies.

In the long term, good nutrition requires reliable access to a variety of fresh foods. Improved food production and distribution systems are needed to provide more food and the right kinds of food. Adding vitamin A to commonly available foods or modifying crops like rice could also improve access. In the short term, our results provide convincing evidence that vitamin A supplements prevent death in childhood. The results are so clear that we believe trials comparing Vitamin A to placebo would no longer be ethical. Instead, further studies should examine questions such as whether more regular doses are more effective than a single, large dose, which is what most of the children received in the trials we reviewed.

In summary, Vitamin A supplements are highly effective and they are cheap to produce and administer, particularly since they can be delivered when children receive other interventions, like vaccines. The WHO should continue its work towards providing Vitamin A supplements to all children who are at risk of deficiency, especially in developing countries."

1. Imdad A, Herzer K, Mayo-Wilson E, Yakoob MY, Bhutta ZA. Vitamin A supplementation for preventing morbidity and mortality in children from 6 months to 5 years of age. *Cochrane Database of Systematic Reviews* 2010, Issue 12. Art. No.: CD008524.